Phase Response

There are three types of *phase response* that a filter can have: **zero phase**, **linear phase**, and **nonlinear phase**. An example of each of these is shown in Figure 19-7. As shown in (a), the *zero phase* filter is characterized by an impulse response that is symmetrical around sample zero. The actual shape doesn't matter, only that the negative numbered samples are a mirror image of the positive numbered samples. When the Fourier transform is taken of this symmetrical waveform, the phase will be entirely zero, as shown in (b).

The disadvantage of the zero phase filter is that it requires the use of negative indexes, which can be inconvenient to work with. The linear phase filter is a way around this. The impulse response in (d) is identical to that shown in (a), except it has been shifted to use only positive numbered samples. The impulse response is still symmetrical between the left and right; however, the location of symmetry has been shifted from zero. This shift results in the phase, (e), being a *straight line*, accounting for the name: *linear phase*. The slope of this straight line is directly proportional to the amount of the shift. Since the shift in the impulse response does nothing but produce an identical shift in the output signal, the linear phase filter is equivalent to the zero phase filter for most purposes.

Figure (g) shows an impulse response that is *not* symmetrical between the left and right. Correspondingly, the phase, (h), is *not* a straight line. In other words, it has a *nonlinear phase*. Don't confuse the terms: *nonlinear and linear phase* with the concept of *system linearity* discussed in Chapter 5. Although both use the word *linear*, they are not related.

Why does anyone care if the phase is linear or not? Figures (c), (f), and (i) show the answer. These are the **pulse responses** of each of the three filters. The pulse response is nothing more than a positive going step response followed by a negative going step response. The pulse response is used here because it displays what happens to both the rising and falling edges in a signal. Here is the important part: zero and linear phase filters have left and right edges that look the *same*, while nonlinear phase filters have left and right edges that look *different*. Many applications cannot tolerate the left and right edges looking different. One example is the display of an oscilloscope, where this difference could be misinterpreted as a feature of the signal being measured. Another example is in video processing. Can you imagine turning on your TV to find the left ear of your favorite actor looking different from his right ear?

It is easy to make an FIR (finite impulse response) filter have a linear phase. This is because the impulse response (filter kernel) is directly *specified* in the design process. Making the filter kernel have left-right symmetry is all that is required. This is not the case with IIR (recursive) filters, since the recursion coefficients are what is specified, not the impulse response. The impulse response of a recursive filter is *not* symmetrical between the left and right, and therefore has a *nonlinear* phase.